Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. Consider the following matrices:

$$A = \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 \\ -4 & 0 \\ 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & -2 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & 1 \\ -2 & 2 & 4 \end{bmatrix}, \quad F = \begin{bmatrix} -2 & 1 & 3 \\ -1 & 1 & -2 \\ 0 & 1 & 3 \end{bmatrix}$$

Determine the following matrices, only if they are defined. If they are not defined, explain why.

A) $A + B$
B) $B + A$
C) $A - B$
D) $D + E$
E) $-A + 2B$
F) $A + C$
G) $A + D$
H) $A - A$
I) $C - C$
J) AB
K) AC
L) CA
M) AD
N) DA
O) EF
P) FE
Q) $A(3D)$
R) $(AC)D$
S) $A(CD)$
T) $(2A)(C(3D))$
U) $(2A)((2C)(2D))$
V) $A(CD + D)$
W) C^3

A) $\begin{bmatrix} 1 & 5 \\ -1 & 3 \\ -1 & -2 \end{bmatrix}$,

B) $\begin{bmatrix} 1 & 5 \\ -1 & 3 \\ -1 & -2 \end{bmatrix}$

C) $\begin{bmatrix} 3 & -1 \\ 7 & 3 \\ -1 & 0 \end{bmatrix}$

D) Not defined, because D is a 2×3 matrix and E is a 3×3 matrix, so they do not have the same dimensions.

E) $-1 \cdot \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ -1 & -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} -1 & 3 \\ -4 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -4 & 5 \\ -11 & 3 \\ 0 & -2 \end{bmatrix}$

F) Not defined, because $A_{3\times 2} = [a_{ij}] = A$ and C is a $C_{2\times 2}$ matrix, so they do not have the same dimensions.

G) Not defined, because A is a 3×2 matrix and D is a 2×3 matrix, so they do not have the same dimensions.

H) $H = 0_{3x3}$

I) $H = 0_{2x2}$